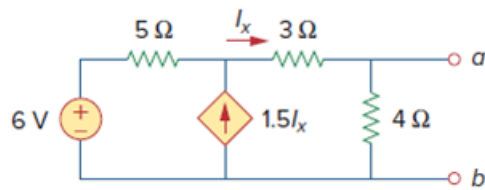


Problem

Find the Thevenin equivalent circuit of the circuit in Fig. 4.34 to the left of the terminals.

Figure. 4.34:



Step-by-step solution

Step 1 of 7

Refer to Figure 4.34 in the textbook.

Redraw the circuit as shown in Figure 1.

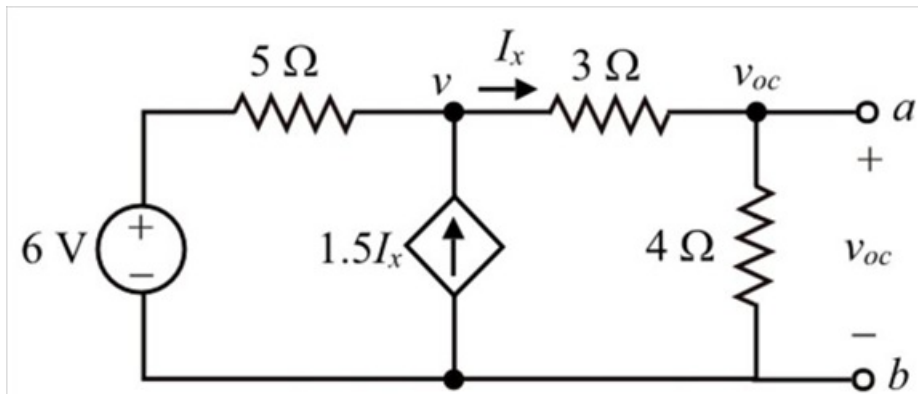


Figure 1

Step 2 of 7

Apply Kirchhoff's current law at node v .

$$\frac{v-6}{5} + I_x = 1.5I_x$$

$$\frac{v-6}{5} = 1.5I_x - I_x$$

$$\frac{v-6}{5} = 0.5I_x \dots\dots (1)$$

From the circuit, observe that the current I_x is as follows:

$$I_x = \frac{v}{3+4}$$

$$= \frac{v}{7} \dots\dots (2)$$

Substitute $\frac{v}{7}$ for I_x in equation (1).

$$\frac{v-6}{5} = 0.5\left(\frac{v}{7}\right)$$

$$v-6 = 0.35714v$$

$$v-0.35714v = 6$$

$$v = \frac{6}{0.64286}$$
$$= 9.3333 \text{ V}$$

Step 3 of 7

Substitute 9.3333 V for v in equation (2).

$$I_x = \frac{9.3333}{7}$$
$$= 1.3333 \text{ A}$$

Since $V_{Th} = v_{oc}$, calculate the Thevenin's voltage.

$$v_{oc} = 4I_x$$

Substitute 1.3333 V for I_x .

$$v_{oc} = 4(1.3333)$$
$$= 5.333 \text{ V}$$

Hence, the Thevenin's voltage is 5.333 V.

Step 4 of 7

Due to the presence of the dependent source, place or excite the network with unity voltage source (1 V voltage source) across the terminals a and b to calculate the Thevenin resistance. Set the independent voltage source to zero. The modified circuit is shown in Figure 2.

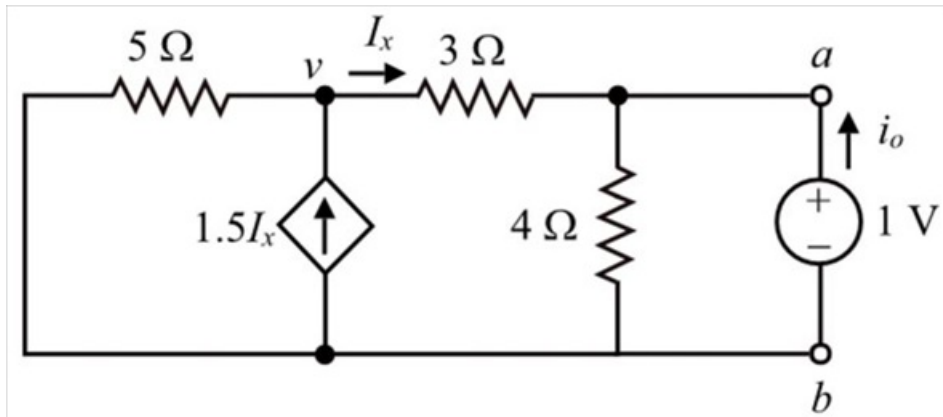


Figure 2

Step 5 of 7

Apply Kirchhoff's current law at node v .

$$1.5I_x = \frac{v}{5} + I_x$$

$$0.5I_x = \frac{v}{5}$$

$$0.5\left(\frac{v-1}{3}\right) = \frac{v}{5}$$

$$0.5\left(\frac{v-1}{3}\right) = \frac{v}{5}$$

$$0.8334v - 0.8334 = v$$

$$v = \frac{0.8334}{-0.1666}$$

$$= -5 \text{ V}$$

Step 6 of 7

Apply Kirchhoff's current law at node a .

$$i_o = \frac{1}{4} + \frac{1-v}{3}$$

$$= \frac{1}{4} + \frac{1-(-5)}{3}$$

$$= 0.25 + 2$$

$$= 2.25 \text{ A}$$

Calculate the Thevenin's resistance.

$$R_{Th} = \frac{1}{i_o}$$

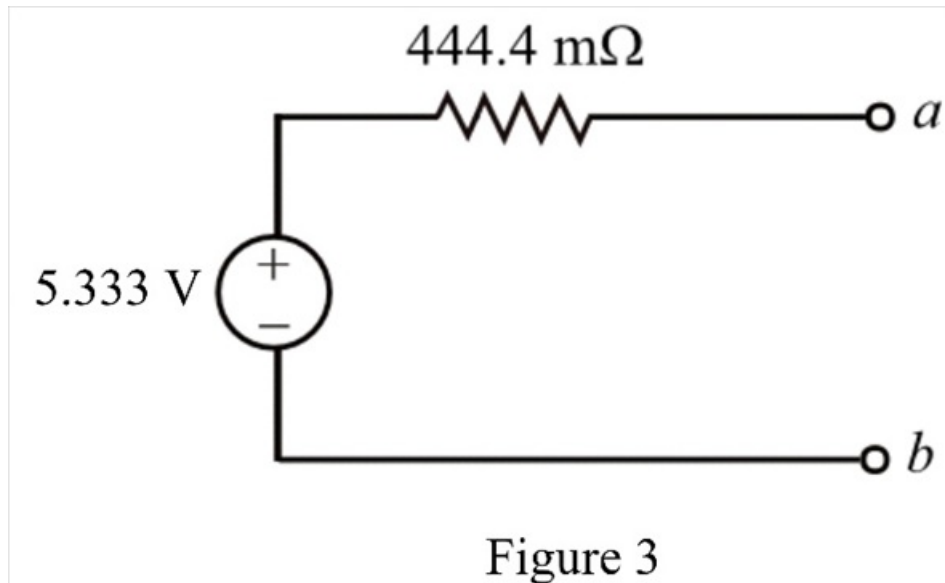
$$= \frac{1}{2.25}$$

$$= 0.4444 \Omega$$

$$= 444.4 \text{ m}\Omega$$

Hence, the Thevenin's resistance is 444.4 mΩ.

The Thevenin equivalent circuit has Thevenin voltage (V_{Th} or v_{oc}) in series with Thevenin resistance (R_{Th}) as shown in Figure 3.



Thus, the Thevenin's equivalent circuit is shown in Figure 3.